INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE FIGHTING, WATER TANKER, 4,000-GALLON CAPACITY

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers a commercial water tanker truck. The tanker has a 6x4 chassis with a diesel engine and an automatic transmission; a two person cab with two doors; a 4,000-gallon water tank; a 1,250-gallon per minute (gpm) fire pump; equipment compartments; and hose. The water tanker truck is intended to store and transport water over paved roads, gravel roads, or unimproved surfaces, and draft, pump, and transfer water in support of brush, structural, and aircraft firefighting vehicles.

2. SALIENT CHARACTERISTICS. The vehicle shall be in accordance with the applicable requirements of the following chapters of National Fire Protection Association (NFPA) 1901, *Standard for Automotive Fire Apparatus*, Current Edition, for Mobile Water Supply Apparatus:

Chapter	Title
1	Administration
2	Referenced Publications
3	Definitions
4	General Requirements
7	Mobile Water Supply Fire Apparatus
12	Chassis and Vehicle Components
13	Low-Voltage Electrical Systems and Warning Devices
14	Driving and Crew Areas
15	Body, Compartments, and Equipment Mounting
16	Fire Pumps and Associated Equipment
19	Water Tanks
23	Line Voltage Electrical Systems

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FSC 4210

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2.1 Administration.

2.1.1 Environmental conditions.

2.1.1.1 Altitude. Unless otherwise specified (see 6.2), the vehicle, including the pumping system, shall be designed for operation at 2,000 feet above sea level.

2.1.1.2 Temperature range. The vehicle shall be capable of satisfactory storage and operation in temperatures ranging from 0 to 125 °F. The vehicle shall be equipped with a winterization system that extends the operational range of the vehicle. The winterization system shall protect the agent pump, agent piping systems, firefighting systems, and water tank to 0 °F. It shall also include the vehicle engine oil and coolant systems. While in operation, the appropriate systems requiring winterization shall be powered by the vehicle. While in storage, the winterization system shall be powered through the electrical shoreline connection (see 2.9.1).

2.1.2 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

2.1.3 Roadability.

2.1.3.1 Operating terrain. The vehicle shall operate on paved and graded gravel roads.

2.1.3.2 Acceleration. The fully loaded vehicle shall accelerate from 0 to 35 miles per hour (mph) within 25 seconds on a level, paved road.

2.1.3.3 Maximum speed. The fully loaded vehicle shall attain a minimum top speed of 50 mph on a level, paved road.

2.1.4 Weight and dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length	390 inches
Width	102 inches (excluding mirrors)
Height	120 inches

2.1.4.1 Capacity. The vehicle shall have a net payload capacity of 4,000 gallons of water plus 500 pounds for crew members and 1,000 pounds of equipment. The chassis will include a heavy-duty driveline assembly (including heavy-duty axles and suspension) rated for “on-highway and off-road” usage consistent with at least the net payload specified above. The suspension system shall include rear axle torque rod/axle configuration which allows for vehicle flexibility to traverse uneven off-road terrain.

2.1.5 Turning diameter. The fully loaded vehicle shall have a wall-to-wall turning diameter of less than three times the length of the vehicle.

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2.1.6 Angles of approach and departure. The fully loaded vehicle shall have angles of approach and departure of not less than 20° as required by NFPA 1906, for use in wildland firefighting operations.

2.1.7 Vehicle Stability. Vehicle shall comply with rollover stability criteria as defined by NFPA 1901.

2.1.8 Manuals.

2.1.8.1 Technical manuals. The overall format for the manuals may be military specification, commercial, or a combination of both. Each technical manual shall have a title page.

a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production inspection. Any changes or corrections noted by the procuring activity shall be corrected and updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity and a technical order (TO) number is assigned, the contractor shall pack two complete sets of both hard copy and compact disc (CD) with each vehicle. One complete set of manuals in rewriteable CD format shall be sent to the procuring activity for stock. The procuring activity's address will be provided.

c. The contractor shall grant the United States (U.S.) Air Force a non-exclusive, non-assignable, royalty-free U.S. Government (Government) Purpose License to scan into CENTRA (the U.S. Air Force repository) and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals is not to be sold, disclosed, or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the vehicles described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO number and publication date, as well as distribution, warning, handling and destruction statements, on the cover / title pages of the TO.

2.1.8.1.1 Operator's manuals. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including the fire-extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include at least the following:

a. Full illustrations and descriptions of the location and function of all controls and instruments.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

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c. Checks and adjustments in preparation for placing the vehicle for service upon receipt from the contractor.

d. Preparation for shipment or storage.

e. Warranty information and period of the warranty for the complete vehicle and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of and step-by-step instructions for the operation of the vehicle and its fire-extinguishing system(s) and auxiliary equipment.

g. Description of the post-operational procedures (draining, flushing, etc.).

h. Checklists for the daily maintenance inspection and mission readiness checks that the operator is expected to perform, including basic troubleshooting procedures.

i. Procedures for towing a disabled vehicle.

j. Procedures for changing a tire.

k. Schedules for required preventative maintenance and required periodic maintenance.

l. A line-art drawing of the vehicle, including front, rear, left, and right side views, showing basic dimensions and weights (total vehicle and individual axle weight for the unloaded and fully loaded vehicle).

m. Procedures and maximum speed limits for towing a trailer, including hazards, risks, and special requirements.

2.1.8.1.2 Service manuals. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance, as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index, as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Firefighting system schematic(s).

b. Hydraulic schematic.

c. Pneumatic schematic.

d. Electrical schematic.

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e. Winterization schematic.

f. Fuel schematic.

g. Schedules for required preventative maintenance and required periodic maintenance.

h. Location, procedure, and interval for parts of the truck and equipment that require lubrication.

2.1.8.1.3 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.1.9 Painting, plating, and corrosion control.

2.1.9.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

2.1.9.2 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. Unless otherwise specified (see 6.2), the exterior finish color shall be Candy Apple Red, Sikkens Color Number FLNA3021, DuPont Color Number 97902U or 4737U, PPG Color Number 71528 (the PPG name for this color is Cardinal Red), or equal. When specified, the exterior finish color shall be Desert Sand, Color Number 30313 of SAE AMS-STD-595. When specified, the exterior finish color shall be Olive Drab with Flat Finish, Color Number 34064 of SAE AMS-STD-595. The interior of all compartments shall be painted with an impact-resistant, textured coating that resists stains, scuffs, chips, and scratches.

a. For vehicles painted Candy Apple Red, the cab upper body (from the bottom of the windshield) and roof shall be painted White, Color Number 17875 of SAE AMS-STD-595. All bright-metal and anodized parts, such as mirrors, horns, light bezels, treadplate, and roll-up compartment doors, shall not be painted. Compartment interiors shall be painted with an impact resistant, textured coating that resists stains, scuffs, chips, and scratches.

b. For vehicles painted Desert Sand or Olive Drab, all exterior surfaces, including all normally bright-metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted body color. This includes fenders, compartment doors, shelves, and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray.

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2.1.9.3 Horizontal reflective striping. Horizontal reflective striping in accordance with NFPA 1901 shall be applied around the vehicle in an approximate plane with the headlights. Bright metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10-inches wide with three reflective stripes (one inch reflective, one inch body color, six inches reflective, one inch body color, and one inch reflective). The reflective stripes shall be white.

2.1.9.4 Chevron Striping. In accordance with NFPA 1901 at least 50 percent of the rear-facing vertical surfaces, visible from the rear of the apparatus, excluding any pump panel areas not covered by a door, shall be equipped with retroreflective striping in a chevron pattern sloping downward and away from the centerline of the vehicle at an angle of 45 degrees. 3M part number 983-17 (red) and 983-23 (florescent yellow/green) reflective diamond grade or equivalent shall be utilized. Each stripe in the chevron shall be a single color alternating between red and fluorescent yellow in accordance with NFPA 1901. Each stripe shall be 6 inches (150 millimeter (mm)) in width in accordance with NFPA 1901 and the example provided below.



2.1.9.5 Lettering. Vehicles painted Candy Apple Red shall have the letters "UNITED STATES" and "AIR FORCE" or any other applicable military service applied in synthetic or encapsulated gold leaf, with outline and black shadow, on the front door on both sides in long radius elliptical arches above and below the lettering center line. The size of the lettering shall be a minimum of 2½ inches to a maximum of 6 inches. The Air Force Fire Emergency Services logo shall be placed

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at a location to be determined during the post-award meeting. Vehicles painted Desert Sand or Olive Drab shall have no lettering or logo.

2.1.10 Identification plate. A permanently marked identification plate shall be securely mounted at the driver's compartment. The identification plate shall contain the following information:

a. Nomenclature.
b. Manufacturer's make and model.
c. Manufacturer's serial number.
d. U.S. Air Force vehicle registration number.
e. National Stock Number (NSN).
f. Vehicle curb weight: pounds (kilograms).
g. Payload, maximum: pounds (kilograms).
h. Gross Vehicle Weight (GVW): pounds (kilograms).
i. Fuel capacity and type: gallons (liters).
j. Date of delivery (month and year).
k. Warranty (months and miles (kilometers)).
l. Contract number.
m. Order number.
n. Paint color and number.
o. Length, width, and height of vehicle in inches and centimeters.

2.1.11 Delivery. All units shall be driven to the receiving installation for CONUS locations or port of call for OCONUS locations.

2.1.12. Warranty. The warranty period shall not begin until training has been completed at the receiving installation.

2.1.13. Air Force registration numbers. The manufacturer shall maintain a list of Air Force registration numbers and corresponding serial numbers for the purpose of identifying vehicles affected by service bulletins.

2.2 General requirements.

2.2.1 Hose storage.

2.2.1.1 Hose storage area. The vehicle body shall have a hose bed, with one or more removable covers, in accordance with NFPA 1901. Hose bed covers will be red for vehicles painted Candy Apple Red, tan for Desert Sand vehicles, and dark green for Olive Drab vehicles. Covers will be secured by snap fasteners.

2.2.1.2 Cross-lay hose beds. The vehicle shall have two cross-lay hose beds between the back of the cab and the front of the water tank, each capable of carrying a 200-foot preconnected 1¾-inch double jacketed fire hose. The hose beds shall be equipped with one or more removable covers. Sufficient clearance shall be provided between the hose and cover to allow a preconnected hose nozzle to be pulled through from either side of the vehicle. Hose bed covers will be red for vehicles

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painted Candy Apple Red, tan for Desert Sand vehicles, and dark green for Olive Drab vehicles. Covers will be secured by snap fasteners.

2.2.1.3 Hose board trays. The vehicle shall be equipped with two running-board hose trays located under discharge panels on both the driver and crew chief side. Each tray shall be equipped with a mat and shall be designed to allow for proper drainage.

2.2.2 Suction hose.

2.2.2.1 Hard suction hoses. Two 10-foot long, lightweight 6-inch hard suction hoses, with 6-inch National Hose thread long-handle female couplers on one end and 6-inch National Hose thread long-handle male couplers on the other, shall be mounted on the vehicle. A suction strainer shall also be mounted on the vehicle. The hose and suction strainer mounting system design shall be subject to approval by the procuring activity.

2.2.2.2 Soft suction hose. The vehicle shall be equipped with a minimum 15-foot long, 6-inch soft suction hose, with a 4½-inch National Hose thread long-handle female coupler on one end and a 6-inch National Hose thread long-handle female coupler on the opposite end. The soft suction hose and couplers shall be stored in a compartment.

2.2.3 Minor equipment.

2.2.3.1 Fire hose and nozzles. The vehicle shall be designed to accommodate all items listed in NFPA 1901. The fire hose and nozzles shall not be provided with the vehicle. The vehicle shall have compartment space for the full complement of equipment listed in NFPA 1901. The self-contained breathing apparatus (SCBA), the spare SCBA cylinders, and the Automatic External Defibrillator shall be provided by the purchaser. Two individual compartments, or a two-bottle rack in a compartment, shall be included for two purchaser-provided one-hour-capacity SCBA air bottles**.**

2.2.3.2 Chocks. Two folding wheel chocks and mounting hardware shall be provided as loose equipment. The wheel chocks shall hold the apparatus, when loaded to its GVWR or GCWR, on a hard surface with a 20-percent grade with the transmission in neutral and the parking brake released.

2.2.4 Portable water-storage pond. The vehicle shall be equipped with a 4,000-gallon-minimum-capacity portable water-storage pond/emergency water-supply storage tank. The pond wall height shall not exceed the height of the bottom of the water-tank dump valve while the vehicle and pond are on a common level surface. The pond shall be stored onboard the vehicle in a rack or compartment in a manner protecting it from UV degradation. The pond shall be capable to be loaded and/or offloaded by no more than 2 crew members from a height of no greater than 48 inches, as measured from the bottom of the pond to the ground. There shall be no requirement to climb or step on to the vehicle to access the pond for the purposes of loading or offloading from the vehicle.

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2.3 Chassis and vehicle components.

2.3.1 Engine. The truck shall have a diesel engine that is certified to comply with the Environmental Protection Agency (EPA) on-highway emission requirements at the time of manufacture. The engine shall run on Ultra-Low-Sulfur Diesel Fuel unless otherwise specified (see 6.2).

2.3.1.1 Fuel filters. Primary and secondary fuel filters and a fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve, and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel-filter elements shall be replaceable without loss of engine prime.

2.3.2 Exhaust system. The exhaust system outlet(s) shall be directed away from personnel accessing equipment compartments or the tank manhole access ladder.

2.3.2.1 Vehicle-mounted diesel-exhaust filter system. A vehicle-mounted, direct-source capture exhaust filter system shall be supplied with the apparatus that shall prevent exposure to, and contamination from, exhaust emissions in addition to the manufacturer's after-treatment device (ATD) in accordance with NFPA 1500. This system shall be installed after the engine manufacturer's ATD and before the diffuser tip in the end-tailpipe. This filter system shall work automatically whenever the apparatus exits and returns to the station. In addition, the system shall have the capability to be used while on-scene, outside the station. The diesel-exhaust-removal system shall travel with the apparatus. The system shall not require building modifications or hanging hoses for the system to operate. The installation of the system shall be performed by the exhaust-removal system manufacturer and shall not be subcontracted. The vehicle-mounted exhaust filter system shall meet all NFPA, National Institute for Occupational Safety and Health (NIOSH), and OSHA standards for preventing exposure to carcinogenic compounds that exist in diesel exhaust. When a standard diesel engine is specified (see 6.2), the vehicle shall be supplied with a direct-source capture system that includes an electronic control module, which engages a diverter unit to direct diesel exhaust from the vehicle's exhaust system into a ceramic filter where diesel particulates and gasses are trapped. When equipped with this system, the vehicle shall be provided with a spare filter upon delivery to the end user.

2.3.3 Brake system. The vehicle shall be equipped with an all-wheel antilock brake system; the brakes shall be fully air-actuated. Brakes shall be in accordance with Code of Federal Regulations (CFR) 49 CFR 393.40 through 393.42(b) and 393.43 through 393.52. The braking system complete with all necessary components shall include:

a. Air compressor having a capacity of not less than 12 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the reservoir tank. Tank drains shall be appropriately labeled.

c. Automatic moisture ejector on air storage reservoir.

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d. Automatic slack adjusters on cam brakes or internal self-adjusting brakes on wedge and disc brakes on all axles.

e. Spring-set parking brakes.

All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake-system components. No part of the braking system shall extend below the bottom of wheel rims, to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

2.3.3.1 Air dryer. A replaceable-cartridge desiccant air dryer shall be installed in the air-brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

2.3.3.2 Compressed-air shoreline. A flush mounted, checked, auto-eject compressed-air shoreline connection shall be provided to maintain brake-system pressure while the vehicle is not running. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left side front corner of the body. An air chuck shall be provided as loose equipment.

2.3.3.3 Auxiliary braking system. The service brakes shall be augmented by one of the following auxiliary braking systems:

a. A system which opens all or some of the engine exhaust valves near the end of the compression stroke, thereby converting vehicle motion to a pumping loss. The engine brake shall be approved by the engine manufacturer.

b. A hydrodynamic retarder integral with the transmission.

c. A controlled gate valve in the exhaust manifold, which produces engine back pressure. The exhaust brake shall be approved by the engine manufacturer.

A dash-mounted switch shall be provided to activate, modulate, or cut out the brake augmentation. The switch shall be marked to indicate its position. When active, the system shall be fully controlled by means of the conventional driving controls to apply retardation during vehicle deceleration, and cut out in the other operating modes.

2.3.4 Tires and wheels. The vehicle shall be equipped with tubeless steel-belted radial tires with highway-type tread rated for specified highway speeds as well as off-road conditions, mounted on steel-disc wheel assemblies. If all tire and wheel assemblies are identical, one spare tire and wheel assembly shall be provided. If two different tire and wheel assemblies are provided, two spare tire and wheel assemblies shall be provided, one of each configuration. Spare tire tread patterns, size,

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and type will be the same as those mounted on the vehicle. The spare tire(s) and rim(s) shall be shipped with the vehicle as loose equipment.

2.3.5 Tire chains. When specified (see 6.2), the vehicle shall be equipped with driver-activated automatic-type tire chains for the rear axles.

2.3.6 Fenders. Rear fenders and fender liners having tire chain clearance shall be provided. For vehicles painted Desert Sand or Olive Drab, the fender flares shall be painted the job color.

2.3.7 Steering. The vehicle shall be equipped with power steering.

2.3.8 Transmission. A fully automatic transmission with a hydraulic torque converter and at least four forward speeds shall be provided. The normal driving-range selector position shall provide at least four gear ratios without movement of the selector. The transmission shall include an oil-cooling system capable of limiting the transmission temperature to the maximum recommended by the transmission manufacturer. The transmission shall have a direct-drive lockup for pumping.

2.3.9 Traction control. The vehicle shall be equipped with limited-slip or automatic locking differentials, including a driver-controlled capability to lock the two tandem axles together for increased rear traction as conditions warrant.

2.3.10 Fuel tank. Fuel tank(s) having a minimum usable capacity in accordance with NFPA 1901 or 50 gallons, whichever is greater, shall be provided. The location of the fuel tank(s) shall protect the tank(s) from mechanical damage during normal use of the vehicle. The tank fill location or fuel tank cap shall be marked “Diesel Fuel Only” or “Ultra-Low Sulfur Fuel Only”.

2.3.11 License-plate bracket. A lighted license-plate bracket shall be provided at the front bumper and left rear of the vehicle.

2.4 Low-voltage electrical systems. The vehicle shall have a 12-volt electrical and starting system.

2.4.1 Alternator. A single alternator charging system in accordance with NFPA 1901 shall be provided. The minimum continuous electrical load shall include operation of the air conditioning system.

2.4.2 Batteries. Batteries shall be of the maintenance-free type. The addition of water shall not be required during the battery’s normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

2.4.2.1 Battery compartment. The batteries shall be enclosed in a ventilated, corrosion-resistant, weather-resistant box or compartment and shall be readily accessible.

2.4.3 Battery charger or conditioner. The vehicle shall have a direct current (DC) taper-type battery charger or an automatic battery conditioner, providing a minimum 12-amp output. The charger/conditioner shall be permanently mounted on the vehicle in a properly ventilated,

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accessible location. The charger/conditioner shall be powered from the electrical shoreline receptacle (see 2.9.1). A charging indicator shall be installed next to the receptacle. When a battery conditioner is provided, the conditioner shall monitor the battery state of charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging.

2.4.4 Warning lights. All warning lights shall use light-emitting diode (LED) elements. The warning-light system, related components, and devices shall be in accordance with NFPA 1901.

2.4.4.1 Light bar. A six-element LED light bar, with both forward and side facing heads, shall be mounted on the cab roof. Forward-facing lenses shall be red-white-red-red-white-red, with the white lights switched off in blocking right-of-way mode. Rearward facing lenses shall be red-amber-red-red-amber-red. The light bar shall be separately switched from the warning-light panel. All trucks shall be equipped with a steady red front warning light as required by California state law regardless of shipping destination.

2.4.4.2 Warning-light color. When specified (see 6.2), the rearward, red lights shall be replaced with amber. When specified (see 6.2), all red warning lights shall be replaced with blue.

2.4.4.3 Headlight flashing system. A high beam, alternating/flashing, LED headlight system shall be provided. The headlight flasher shall be separately switched from the warning-light panel.

2.4.5 Audible warning devices.

2.4.5.1 Siren. The vehicle shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise-canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and right front crew member. Siren-activating foot switches shall be located in front of the driver and the right front crew member. The siren speaker shall be rated at 100 watts minimum and shall be located in a guarded position as low and as far forward on the vehicle as practical.

2.4.5.2 Horn. Dual forward-facing air horns shall be installed in protected locations near the front of the truck. The air horns shall be activated by foot switches located in front of the driver and the officer and by a switch located at the pump panel.

2.4.6 Work lighting.

2.4.6.1 Cab interior lights. Cab interior light levels shall be a minimum of 2 foot-candles.

2.4.6.2 Compartment lights. White LED lighting sufficient to provide an average minimum illumination of 2.0 foot-candles shall be provided in each compartment greater than 4.0 cubic feet and having an opening greater than 144 square inches. Where a shelf is provided, this illumination shall be provided both above and below the shelf. Lights shall automatically illuminate only when the respective door is opened and the vehicle electrical master switch is in the "on" position.

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2.4.6.3 Ladder, step, and area lights. Non-glare white LED lighting shall be provided at ladders and access steps where personnel work or climb during night operations. These area lights shall be controlled with switches on the cab instrument panel and near the light sources.

2.4.6.4 Spotlights. Two LED spotlights, with a minimum rating of 12,000 lumens each, shall be mounted on the cab roof and controlled from inside the cab.

2.4.7 Scene lights. A total of eight, 12-volt LED high-mounted floodlights, with a minimum rating of 12,000 lumens each, shall be provided to illuminate the work areas around the vehicle: two on each side, two at the rear, and two in the front. Individual switches shall be located in the work areas and on the instrument panel. A switch shall be provided to control the lights on the left side of the vehicle, a second switch shall be provided to control the lights on the right side of the vehicle, and a third switch shall be provided to control the lights at the rear of the vehicle. A separate switch for the front lights shall be located on the instrument panel only. Three-way switches shall be used.

2.4.8 Radio circuit. The vehicle shall have two separate 30-amp circuits, with breakers and at least 6 feet of wire routed to a space provided adjacent to the driver and officer for purchaser provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose.

2.4.9 Radio Antennas. An antenna mounting base with coaxial cable and waterproof cap shall be provided for 2-way radios. The mount shall be located on the cab roof to the rear of the front passenger's seat and will be clear of any obstructions. The cable shall be routed from the roof to the center console; length of the cable shall be determined at Post Award.

2.4.10 Auxiliary power receptacles. The vehicle shall have two 12-volt auxiliary power receptacles mounted adjacent to the driver and crew member positions, preferably in the instrument panel or in a center radio-mounting console.

2.5 Driving and crew areas.

2.5.1 Cab. The vehicle shall have a two-person, two-door, non-tilting cab. A tilt steering column shall be provided. A vehicle data recorder shall be provided in accordance with NFPA 1901. Provisions must be made for fire helmet storage for all seated positions in accordance with NFPA 1901.

2.5.1.1 Seats. The truck shall have seats in accordance with NFPA 1901. The driver's seat shall be of an adjustable air-suspension design. The crewmember seats shall be of a non-suspension design. Each crewmember seat (excluding the drive position) shall have a backrest, brackets, and pad covers designed to store a one-hour-capacity SCBA. All SCBA-type seats in the cab shall have a "Hands-Free" auto-clamp-style bracket in its backrest. The bracket shall include an automatic spring clamp that allows the occupant to store the SCBA bottle by pushing it into the seat back. In the event of an accident, the inertial components within the clamp shall constrain the SCBA bottle in the seat and shall exceed the NFPA standard of 9G. Bracket designs shall not have manual restraints (belts, straps, or buckles) that could be inadvertently left unlocked and allow the SCBA to move freely within the cab during an accident. Each seat shall be provided with a Type 2 seatbelt

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assembly (i.e., 3-point restraint) in accordance with 49 CFR 571.209. All seats shall be equipped with a seatbelt sensor in compliance with NFPA 1901. All seatbelts shall be red in color.

2.5.1.2 Cab interior sound level. The maximum sound level at any seat location shall not exceed 90 dBA without any warning devices in operation, as measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

2.5.1.3 Windshield and windows. The windshield and windows shall be of tinted safety glass in accordance with industry standards.

2.5.1.4 Heated Mirrors. Outside rearview mirrors shall be installed on each side of the cab, mounted on fold-back brackets. The mirrors shall also have electrically heated heads. Mirror heating controls shall be located on the instrument panel within reach of the seated driver.

2.5.1.5 Climate control system. The manufacturer's standard heater/defroster and air conditioning system shall be provided.

2.5.1.6 Instruments and controls. Gauges shall be provided for oil pressure, coolant temperature, and automatic transmission temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by 14.3.6 of NFPA 1901, the following shall be provided within convenient reach of the seated driver:

a. Master warning-light control switch.

b. Work light switch(es).

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the parking brakes are released or the transmission is in any position other than neutral.

d. An engine-hour meter.

e. Lateral G-Force indicator.

2.5.2 Back-up camera. A hard-wired, rear-vision camera and in-cab monitor shall be provided to aid the driver in safely backing up the vehicle. The monitor shall be located to the right side of the driver. A switch shall be provided to allow the driver to manually activate the back-up camera from within the cab. The back-up camera shall also be switched "on" automatically whenever the vehicle is in the reverse mode of operation.

2.5.3 Active air-purification system. The vehicle shall be equipped with an active air-purification system to provide purification of the air and surfaces inside the cab. The system shall contain a capability to generate hydrogen peroxide (H_2O_2) and reactive oxygen species to sanitize against various microbial species. The system shall produce H_2O_2 at a minimum of 0.02 parts per million

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(ppm) to sanitize the atmosphere inside the apparatus. The H_2O_2 concentration shall not exceed OSHA recommendations. The system shall be sized per application to support virus and bacteria kill rates. The system shall be stand-alone and contain its own airflow mechanism, with a stainless-steel outer housing. The system shall be environmentally friendly and not emit direct UVC outside of the unit. The system shall be resistant to dust particles normally found in apparatus. The working temperature of the system shall be -22 to 149 ºF (-30 to 65 ºC). The system shall be permitted to operate at any time with or without occupants in the cab and shall pose no harm to the occupants from H_2O_2, ozone, or UVC light. The system shall be powered and operating at all times, both in the fire station and during all vehicle operations.

2.5.3.1 Air-purification system certification. The air-purification system shall be third-party tested to verify H_2O_2 production at 0.02 ppm and to support virus and bacteria kill rates. The manufacturer must be ISO 9001:2015 certified and an EPA-registered establishment. The system shall meet all applicable sections of IEC 61373:2010 for shock and vibration and SAE J1455 for electrical specifications. The third-party testing must be performed for effectiveness against SARS-CoV-2 in a chamber at least 1,280 cubic feet in size.

2.6 Body, compartments, and equipment mounting.

2.6.1 Body. The vehicle shall have a body constructed entirely of non-corroding materials. The body shall be at least as strong as if it were constructed of 0.125-inch thick aluminum. A copy of the fire-apparatus manufacturer's warranty shall be included with the bid. The warranty shall state that the body shall be free of structural failures caused by defective design or workmanship for a warranty period of fifteen (15) years from the date the new vehicle is first delivered or 100,000 miles, whichever occurs first, and that defective parts under the warranty shall be repaired or replaced without charge to the original purchaser.

2.6.2 Compartments. The vehicle body shall include sufficient compartment space to hold all of the equipment listed in 2.2.3.

2.6.2.1 Compartment doors. Compartments shall have clear anodized aluminum, counterbalanced, non-locking, roll-up doors and will be provided with an aluminum trey/guard that protects the roll-up door from damage. Door-latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the rolled-up door height exceeds six feet above the ground.

2.6.2.2 Scuffplates. Replaceable scuffplates shall be provided at each compartment threshold to prevent body damage from sliding equipment in and out of the compartments. The scuffplates shall be securely attached to the compartment threshold but shall be easily replaceable in the event of damage.

2.6.2.3 Drip rails. Drip rails shall be provided over each compartment door. If the drip rails are not integral with the body, they shall be of anodized extruded aluminum and shall have a bright finish for vehicles painted Candy Apple Red.

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2.6.2.4 Shelves. An adjustable and removable compartment shelf shall be provided for every 18 inches of each vertical storage compartment door opening. Shelving adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Shelves shall support a minimum of 500 pounds without permanent deformation. Each shelf shall be accessible to crew members standing on the ground. A pull-out and tip-down configuration shall be used for shelving over 54 inches from the ground when the compartment is more than 18 inches deep. Each shelf shall have ¼-inch diameter drain holes located in each corner of the shelves so as to allow for drainage of any water from the stowed equipment.

2.6.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable black mat designed to allow for drainage of any water from the stowed equipment.

2.6.3 Ladder, handrails, and walkways. The vehicle shall have a ladder and walkway to access the manhole(s) and servicing points along the top of the water tank. Folding handrails or tie-on points shall be provided to restrain personnel from falling to the ground. All ladders, stepping, standing, and walking surfaces shall be in accordance with NFPA 1901. Handrails shall be provided in accordance with NFPA 1901.

2.7 Fire pump and associated equipment.

2.7.1 Fire pump. The fire pump shall be a single-stage centrifugal design, with a thick-walled cast-iron pump, a bronze impeller and a stainless-steel shaft, capable of providing 1,250 gpm of water at 150 psi. The cast-iron pump body must not contribute in any way to the premature failure of the fire pump. The pump shall be driven from the chassis-propulsion engine through a split-shaft power takeoff with ratings to handle the full torque and power applied.

2.7.2 Piping and associated components. All metallic surfaces of the piping and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The discharge piping shall flow water at a minimum of 1,250 gpm.

2.7.3 Intake connections. The vehicle shall have two 6-inch intake connections, one on each side near the front of the tank. The vehicle shall have two gated 2½-inch intake connections, one on each side of the vehicle, adjacent to the 6-inch intake connections and fitted with 30° or 45° turn-down fittings. Each intake connection shall be gated and shall have National Hose threads.

2.7.4 Discharge connections. The vehicle shall have two 4-inch minimum discharge connections, one on the right side of the vehicle near the front of the tank and one at the rear of the vehicle on the driver's side. A 4-inch discharge shall not be provided at the pump operator's control panel on the left side of the truck. The vehicle shall have two 2½-inch discharge connections, one on each side of the vehicle near the front of the tank. Each 2½-inch discharge connection shall be equipped with no less than 3-inch full-flow piping and valve with a 2½-inch adapter. Each side discharge connection shall be fitted with a 30° or 45° turn-down fitting. The vehicle shall be equipped with two 2-inch pre-connected swivel discharge outlets to accommodate the cross-lay hoses. All discharge connections shall have National Hose threads.

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2.7.5 Pump-operator control panel. A pump-operator control panel in accordance with NFPA 1901 shall be mounted on the left-center side of the vehicle. Gauges shall be provided. The panel shall have master intake and discharge pressure gauges, not less than 6-inch diameter, and individual discharge outlet pressure gauges, not less than 2-inch diameter. Each discharge outlet shall be controlled from the pump-operator panel through a manual valve having locking features. The water-tank level indicators shall be of the five-light type, one on each side of the vehicle, located at the pump panels; that is, the indicator shall have lights to show the water tank level at quarter-tank-capacity intervals and have a flashing "refill" or "tank empty" light. Discharge and intake controls shall be color coded in accordance with NFPA 1901. A green "battery on" pilot light shall be mounted on the pump-operator control panel.

2.8 Water tank. The vehicle shall have a rectangular water tank with a manufacturer's certified capacity of at least 4,000 gallons. The water tank shall have recessed lids with extrusion welds.

2.8.1 Water tank material. The water tank shall be constructed of UV-resistant polypropylene and shall have a lifetime warranty.

2.8.2 Water-tank baffling. The water tank shall be designed for dynamic baffling in accordance with NFPA 1901.

2.8.3 Tank-to-pump intake line. All metallic surfaces of the tank-to-pump intake line and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The tank-to-pump intake line shall be capable of delivering water at a flow rate in accordance with NFPA 1901 for the certified capacity of the water tank provided. The tank-to-pump valve(s) shall be air or electrically actuated with a manual override.

2.8.4 Fill tower. The water tank shall have a fill tower of not-less-than 20 inches inside diameter for personnel access to the interior of the tank for cleaning and inspection.

2.8.5 Tank fill connection. A 6-inch, slow-open, gated fill connection with 5-inch Storz adaptor shall be provided at the right rear of the water tank.

2.8.6 Dump valves. Pneumatic dump valves shall be installed at the rear face and rear left and rear right sides of the water tank for filling the portable pond provided with the vehicle. The dump valve shall empty 90% of the water tank volume within 3 minutes with the vehicle on a level surface. A rigid folding chute or flexible tube extension shall be provided to drain the water in the tank into the portable pond. The chute or extension shall be stored in a readily accessible location and shall be easily attached to the dump outlet. Controls for the dump valve shall be installed both near the outlet and in the cab in a location readily accessible to the seated driver.

2.9 Line voltage electrical system.

2.9.1 Electrical shoreline connection. The battery charger/conditioner shall be powered from a covered, three-wire, straight-blade, polarized, insulated, labeled, recessed, 120-volt, NEMA 5-20P male plug inlet, in a weatherproof alternating current (AC) auto-eject receptacle. It shall be located

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on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left-side front corner of the body. The shoreline connection's location shall not pose a tripping hazard to pedestrians walking by the vehicle when the shoreline is connected. A weatherproof charge meter shall be installed next to the receptacle. A 50-foot long, three-wire, 20-amp rated, 120-volt, AC power cable, with straight blade (non twist-lock), NEMA 5-20R and 5-20P style, in accordance with NEMA WD-6, connectors, shall be provided. When equipped with a winterization system, the winterization system shall have a labeled activation switch with a pilot light. When specified (see 6.2), the receptacle(s) and power cable(s) shall be for 220 volts in lieu of 110 volts.

2.9.2 Idle reduction technology. An IRT system shall be supplied with the apparatus that shall shut down the main chassis diesel engine during operations not requiring it. This system shall be automated and shall not require intervention from the vehicle operator. The chassis voltage system shall be protected against extreme drain of the battery bank. If the vehicle's voltage drops too low, the automatic engine controls shall start the chassis diesel engine to provide a charge. The IRT system shall power an in-cab climate-control system with a minimum of 13,500 British thermal unit (BTU) A/C evaporator, 300 CFM air flow, and 5,600 BTU heating element. The IRT system shall be available when the chassis air brake is set and when the power take-off (PTO) is not engaged. The system shall interface with the chassis batteries and an auto eject electric shoreline connection. Easy to access control panel(s) and fuse boxes shall be installed on the vehicle. The fuel supply for the IRT shall be the chassis fuel tank. At a minimum the following systems shall be operable when the IRT is engaged: in-cab climate-control system, radios, DOT truck lighting, NFPA-required emergency lights, step lights, and compartment lights.

2.10 Trailer towing package. The vehicle shall be equipped with a square sleeve-type receiver used to tow trailers equipped with a ball tow hitch and other ancillary firefighting equipment. The tow hitch and receiver shall be rated at the maximum inherent towing capability of the vehicle such that the safety and reliability of the vehicle are not degraded. The trailer hitch shall comply with NFPA 1901. The maximum towing capacity and tongue-weight capacity of the vehicle shall be clearly and visibly posted both near the tow-hitch receiver and inside the cab of the vehicle. The manufacturer shall describe any hazards, risks, maximum suggested speed limits or restrictions, and special-use requirements during the solicitation process. This information shall also be listed near the tow-hitch receiver, inside of the cab of the vehicle, as well as in the operator's manual. A brake controller shall be provided for use with a trailer which has an electric braking system. Gladhands in accordance with SAE J318 shall be provided for use with a trailer which has an air braking system.

2.10.1 Safety-chain attachment points. The vehicle shall be equipped with two safety-chain attachment points, one located on each side of the receiver sleeve, symmetrical about the receiver sleeve. The vertical centerline of the safety-chain attachment points shall be located at a parallel distance of not greater than 6 inches to the left and right of the truck centerline. Safety chains shall comply with NFPA 1901 and shall have the ability for standard safety-chain hooks to be attached. The inside radius of the safety-chain hook is between 1 inch and 3 inches.

2.10.2 Electrical-harness receptacle. The vehicle shall have a weatherproof 7-way round (pin) electrical receptacle installed within 12 inches of the receiver sleeve. The electrical receptacle shall have provisions to provide electrical power to a trailer with electric brakes. All terminals of the receptacle shall be wired in accordance with SAE J560. A weatherproof 7-way flat (blade)

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electrical receptacle shall be included as loose equipment. A heavy-duty flasher designed for trailer applications shall be provided to support the additional electrical load of the trailer's lighting system.

2.11 Workmanship. The vehicle, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners; etc. Wire ties may be used for bundling vehicle wiring, but not for support of wire bundles. Support must be provided by insulated metal clamps. The vehicle shall be thoroughly cleaned of all foreign matter.

2.12 Training. A minimum of three days of training shall be provided at the receiving base following the delivery of the vehicle. Training shall consist of driver/operator training, and informational overview training for vehicle maintenance personnel. One three-day training session shall be required at each location, regardless of the number of vehicles shipped to that location.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous lifecycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired a) competitively within a reasonable timeframe, b) meeting appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First production inspection. The first production vehicle shall be subjected to the examinations and tests described in 4.6.3.1 through 4.6.3.6. The contractor shall provide or arrange for all test equipment and facilities.

4.3 Conformance inspection. Each production vehicle shall be subjected to the examinations and tests described in 4.6.3.1, 4.6.3.2.1 through 4.6.3.2.4, 4.6.3.3, 4.6.3.5, and 4.6.3.6.

4.4 Product conformance. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.5 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the manufacturer to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The manufacturer shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID**;** a paragraph-by-paragraph response to the salient characteristics section of this CID shall be provided**.** The response to each paragraph shall clearly state “Fully Comply” or “Bid with Exception” in addition to other data the contractor wishes to submit. The proposal shall be provided with the pricing submission. Failure to provide this information may deem a vendor as non-responsive, and their proposal may be rejected. The manufacturer shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The manufacturer shall identify all modifications made to their commercial model in order to comply with the requirements herein.

4.6 Inspection requirements.

4.6.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.6.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

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e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering and braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

k. Failure of the firefighting system.

4.6.3 Detailed inspection requirements.

4.6.3.1 Examination of product. Each vehicle shall be examined to verify compliance with the requirements herein. A Government-generated checklist that identifies each relevant requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants, and their application, welding, fastening, and markings. Proper operation of each vehicle function shall be verified. Each production vehicle shall be inspected to a reduced version of the checklist.

4.6.3.2 Road tests. The following tests shall be performed in accordance with NFPA 1901 and shall include traversing 5 miles of gravel roads.

4.6.3.2.1 Acceleration test. Each vehicle shall be tested to demonstrate compliance with 2.1.3.2. For the first production truck, a time-distance recorder shall be used to record data for this test.

4.6.3.2.2 Maximum speed test. Each vehicle shall be tested to demonstrate compliance with 2.1.3.3. For the first production unit, a time-distance recorder shall be used to record data for this test.

4.6.3.2.3 Auxiliary braking system test. Each vehicle shall be tested in accordance with NFPA 1901.

4.6.3.2.4 Service brake system test. Each vehicle shall be tested in accordance with NFPA 1901. For the first production vehicle, a time-distance recorder shall be used to record data for this test.

4.6.3.2.5 Turning diameter test. The fully loaded first production vehicle shall be tested to demonstrate compliance with 2.1.5. This test shall be performed on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the vehicle being tested.

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4.6.3.2.6 Mobility test. The fully loaded first production vehicle shall be driven over 10 miles of paved roads and five miles of gravel roads. All loads shall be removed, and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

4.6.3.2.7 Weights and Measurements. The vehicle shall be weighed in both the empty and full configuration. Length, width, and height measurements shall be validated in accordance with 2.1.4.

4.6.3.2.8 Ancillary Equipment. Each vehicle shall be inventoried to ensure that equipment listed in NFPA 1901 is provided. Equipment shall not be mounted to the vehicle. It will be inventoried, packaged, and shipped as loose equipment.

4.6.3.3 Low-voltage electrical system performance tests. Each vehicle shall be tested in accordance with NFPA 1901.

4.6.3.4 Cab interior sound-level test. The cab interior sound levels of the first production vehicle shall be measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

4.6.3.5 Test of fire pump and associated equipment. Each vehicle shall be tested in accordance with NFPA 1901. The costs associated with the independent certification shall be included in the price of the vehicle.

4.6.3.6 Line voltage electrical system testing. Each vehicle shall be tested in accordance with NFPA 1901.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Government documents. Copies of specifications, standards, and handbooks are available online at https://quicksearch.dla.mil/.

6.1.2 Code of Federal Regulations (CFR). CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.

6.1.3 Environmental Protection Agency (EPA) documents. Copies of EPA documents may be obtained from US EPA National Headquarters Building, 1200 Pennsylvania Avenue NW, Washington, DC 20004, or online at https://www.epa.gov/.

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6.1.4 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

6.1.5 International Electrotechnical Commission (IEC) standards. Copies of IEC standards may be obtained from IEC Secretariat, 3 rue de Varemba, PO Box 131, CH-1211 Geneva 20, Switzerland, or online at https://www.iec.ch.

6.1.6 ISO standards. Copies of ISO standards may be obtained from Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, or online at https://www.iso.org/.

6.1.7 National Electrical Manufactures Association (NEMA) documents. Copies of NEMA documents may be obtained from the addressed to National Electrical Manufactures Association, 1300 North 17th Street, Suite 900, Arlington, VA 22209, or online at https://www.nema.org/.

6.1.8 National Fire Protection Association (NFPA) documents. Copies of NFPA documents may be obtained from NFPA, 1 Batterymarch Park, Quincy, MA 02269-9101, or online at https://www.nfpa.org/.

6.1.9 Occupational Safety and Health Association (OSHA) standards. Copies of OSHA standards may be obtained from OSHA, 200 Constitution Avenue, Washington, D.C. 20210, or online at https://www.osha.gov/.

6.1.10 SAE International (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at https://www.sae.org/.

6.2 Ordering data. The contract or order should specify the following:

a. If vehicle will be operated at altitudes greater than 2,000 feet above sea level (see 2.1.1.1).

b. Finish color required (Desert Sand or Olive Drab, in place of Candy Apple Red) (see 2.1.9.2).

c. If diesel fuel is required in lieu of ultra-low-sulfur diesel fuel(see 2.3.2)

d. If driver-activated automatic type tire chains are required (see 2.3.6).

e. Emergency warning-light color required (amber or blue in place of red) (see 2.4.4.2) (NOTE: Applies to USAFE ONLY).

f. Electrical shoreline voltage, 220 volts instead of 110 volts (see 2.9.1) (NOTE: Applies to some USAFE and South West Asia (CENTAF) locations only).

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6.3 Key Words.

Brush
Gravel roads
Paved roads
Unimproved surfaces

MILITARY INTERESTS

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

Agent:
Air Force – 110

Project 4210-2022-003

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